

To editors

Book proposal

The authors have described this book to me and asked whether I would contribute to it. I write to support the proposal and to confirm that I intend to take part.

My work is the determination of losses and efficiency in electrical machines and in converter-fed drives, including the calorimetric methods used where electrical ones are not accurate enough. That is an unusual position from which to judge a design book, but I believe it is a useful one, because measurement is where a design book's claims are either confirmed or quietly avoided.

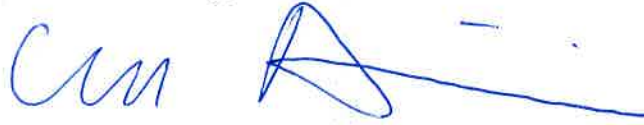
The difficulty is the following. The standard procedures for determining efficiency were written for low-to-moderate frequency machines and, more recently, for converter-fed machines at conventional speeds. They do not transfer to a machine turning at 30 000 r/min on a converter supply. Input-output measurement requires the difference between two large quantities, and at high efficiency with distorted waveforms the uncertainty of that difference approaches the loss being measured. The bandwidth and the phase accuracy of the instrumentation stop being details of the method and become part of the result. Loss segregation, the alternative, assumes that the components can be separated, whereas at kilohertz fundamental frequencies the converter-induced rotor loss, the core loss and the aerodynamic loss are entangled in a way the classical procedures do not anticipate. Calorimetry answers the question honestly, but it is practised in few laboratories and described in fewer books.

The consequence reaches further than the laboratory. Published efficiency figures for high-speed machines are frequently not comparable with one another, because they were obtained by different methods under conditions that were not stated. That bears directly on this book. One of its chapters calibrates the classification criteria for high-speed machines against a population of built and published designs, and such a survey can only be as good as the comparability of what has been published. A book that is explicit about how its own numbers were obtained, and that describes the test rig, the instrumentation and the loss segregation in enough detail for the work to be repeated, is therefore worth more than its page count suggests. Very few design books do this.

Subject to the publishing contract, I intend to contribute the section on converter harmonics and the rotor heating they cause, in the chapter on the drive, and the section on the test rig, instrumentation and loss segregation at high speed, in the industrial case-study chapter. I have worked on the

measurement of losses and efficiency in electrical machines and drives over 20 years, including many academic and industrial project as well as participated in IEC and CENELEC standardisation committees, and would bring these works to the book.

Yours sincerely,

A handwritten signature in blue ink, consisting of a stylized 'L' and 'A' followed by a long horizontal stroke.

Lassi Aarniovuori

Professor (tenured)
Electrical engineering